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RESEARCH ARTICLE

Understanding Customer Evaluations and Human-AI Involvement in Customer-Facing AI Applications: An Exploratory Scoping Review

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ABSTRACT:

As Artificial intelligence (AI) becomes increasingly integrated into customer-facing services, understanding how customers evaluate these applications requires attention not only to their functionality but also to how customers interpret their capabilities and the respective contributions of human and AI actors. In this article, trust and perceived value are positioned as focal dimensions of customer evaluation, whereas human-likeness, anthropomorphism, authenticity, and intentionality offer interconnected yet analytically different perspectives on how customers interpret AI applications and their contribution. However, these concepts are not consistently examined together, and the evidence linking them to customer evaluations remains fragmented across service contexts. This study addresses this gap through an exploratory scoping review guided by two research questions: What does the retrieved literature report about customer evaluations of customer-facing AI applications, particularly regarding trust and perceived value? How does the retrieved literature address human and AI involvement in ways relevant to understanding these evaluations? The review identifies 11 peer-reviewed studies spanning diverse applications and service settings. The findings indicate that customer evaluations vary across AI applications, service contexts, design characteristics, and the ways AI contributes to service outputs. Human-associated characteristics were linked to different customer responses, rather than to uniformly favorable evaluations. AI-generated content and disclosure of AI involvement were associated with less favorable evaluations in some contexts, although the findings did not establish a general negative effect of AI involvement. Regarding human and AI involvement, human-likeness and anthropomorphism received comparatively greater empirical attention than creativity, authenticity, and intentionality as distinct constructs. The review contributes a conceptual map that connects customer evaluations with interpretations of human and AI involvement. Building on this synthesis, the study develops conditional design propositions for customer-facing AI services and identifies directions for future research, emphasizing the context-dependent nature of the evidence.

Keywords: Customer-Facing AI; Customer Evaluation; Human–AI Involvement; Scoping Review

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Introduction

Artificial intelligence (AI) is increasingly embedded in customer-facing services to support service creation, delivery, and interaction, with its role varying according to the nature of the task and the value sought by providers and customers [1]. Recent developments in generative AI have expanded these possibilities by enabling systems to generate text, images, and other forms of content across a wide range of applications [2]. The technologies associated with these developments operate at different levels: generative AI denotes a category of capabilities; large language models provide language-processing and content-generation capabilities; ChatGPT is an application built on such models; and chatbots are conversational interfaces that may rely on different underlying technologies. In this paper, the term customer-facing AI applications serves as the umbrella term for AI technologies whose functions, interactions, or outputs are encountered by customers in service contexts. These applications may mediate interactions, support service processes, or generate customer-facing outputs, but they differ in their functions and roles within service provision.

Artificial creativity provides the conceptual starting point for examining the interplay between human and AI actors in customer-facing AI applications. Creativity is conventionally defined in terms of originality and effectiveness [3]. However, AI's capacity to generate outputs that appear creative challenges established distinctions between human and machine production, particularly when such outputs do not necessarily exhibit the characteristics traditionally associated with human creativity [4]. Although artificial creativity provides a useful conceptual point of departure, the broader question for customer-facing services concerns how customers evaluate AI applications and how their understanding of human and AI involvement shapes those evaluations. In particular, when AI applications mediate interactions, support service processes, or generate outputs, questions arise about how the respective roles, characteristics, and contributions of human and AI actors are understood.

In service contexts, customers evaluate AI based on the systems they interact with, the outcomes they receive, and their overall service experiences. In digital environments, trust can involve beliefs about the competence, integrity, and benevolence of an exchange partner, as well as trust in the technological environment through which the exchange occurs [5]. This distinction becomes more complex when AI occupies an active role in service delivery. Customers may interact with an AI system while simultaneously evaluating the organization that deploys it, meaning that technological performance, organizational responsibility, and the reliability of the resulting service can become intertwined. Perceived value concerns a different dimension of evaluation. Value judgments reflect customers' evaluations of benefits relative to sacrifices, and those judgments can incorporate both functional and experiential aspects of an offering [6]. In customer-facing AI services, customer evaluations may reflect the benefits and costs associated with the technology, the characteristics of the resulting service, and the experience of interacting with or receiving outputs from AI applications. Together, trust and perceived value provide complementary perspectives that position customer evaluation as a relevant lens for understanding how customers respond to AI in service contexts.

Beyond the functions of AI applications and the outcomes they deliver, customer evaluations may also be shaped by how customers understand the processes through which service outcomes are produced. Research on human–AI collaboration emphasizes that outcomes may reflect complementary contributions from human and AI actors rather than the replacement of one by the other [7, 8]. However, in service contexts, customers may have limited visibility into the respective contributions of AI systems and human employees. Consequently, customers' perceptions of the production process may become part of the context in which customer evaluations are formed [9, 10].

Although the aforementioned concepts are connected, existing research has examined customer evaluations of AI and the role of human and AI actors from different conceptual perspectives, resulting in a fragmented understanding of their interrelationships. This creates a need to synthesize how these perspectives relate to one another. In response, the study adopts an exploratory scoping review to map key themes and synthesize emerging patterns in the literature. The study positions customer evaluations as its primary analytical focus, while treating human and AI involvement as a secondary analytical perspective.

The following research questions guide the study:

- What does the retrieved literature report about customer evaluations of customer-facing AI applications, particularly regarding trust and perceived value?
- How does the retrieved literature address human and AI involvement in ways relevant to understanding these evaluations?

By addressing these questions, the study provides a bounded synthesis of reported customer evaluations in relation to representations and interpretations of human and AI involvement. Building on this synthesis, the study develops a set of conditional design propositions for customer-facing AI services while emphasizing the context-dependent nature of the underlying evidence.

1. Conceptual Background and Literature Review

1.1. Customer Evaluations and Perceptions of Production

For the present review, perceptions of production are considered in relation to customer evaluations. Trust and perceived value represent focal evaluative dimensions of customer evaluation, whereas the effort heuristic and the labor illusion may inform those evaluations. These concepts provide a conceptual bridge between customers' perceptions of the production process and their evaluations of customer-facing AI applications.

1.1.1. Customer Trust and Perceived Value

Customer evaluation in service contexts can concern multiple aspects of an application, interaction, or outcome. Trust and perceived value are particularly useful focal concepts because they capture distinct but complementary dimensions of the customer evaluation. Glikson and Woolley [11] synthesized empirical research on human trust in AI, arguing

that trust is influenced by factors associated with the technology, the task, and the context in which AI is used. They identify AI representation and perceived machine intelligence as key antecedents of trust and distinguish between cognitive and affective dimensions of trust. Similarly, Akdim and Casaló [12] examined perceived value in the context of AI-based recommendation services delivered via voice assistants, considering perceived benefits (e.g., convenience, compatibility, and personalization) and perceived costs (e.g., cognitive effort and intrusiveness). Their work situates AI service evaluation within a broader understanding of the value customers perceive from the service, rather than treating the presence of AI as sufficient to determine that evaluation.

Although trust and perceived value may be related, they should not be treated as equivalent indicators of favorable customer responses. Trust concerns customers' willingness to rely on an AI system under conditions of uncertainty, whereas perceived value reflects their overall assessment of the benefits and costs associated with a service. These perspectives indicate that customer evaluations extend beyond assessments of an output's immediate usefulness, as they may also reflect how customers understand and experience the service through which that output is delivered.

1.1.2. Effort Heuristic and Labor Illusion

Perceptions of the production process offer a potential explanation for how customer evaluations are formed. Kruger et al. [10] show that the effort heuristic can serve as an inferential cue for evaluating an outcome, particularly when its quality is difficult to assess directly. They found that participants assigned higher ratings of quality, value, and liking to objects when they believed that more time and effort had been invested in producing them. A related yet distinct perspective is provided by Buell and Norton [9] through the concept of the labor illusion, which highlights how the visibility of service work can shape perceived value even when the service outcome remains unchanged. Their research suggests that making operational processes visible can influence how customers evaluate a service by drawing attention to the work involved in its delivery.

The effort heuristic and the labor illusion both highlight the relevance of production-related information, but they emphasize different aspects of evaluation. The effort heuristic concerns the use of perceived effort as a cue for judging an outcome, whereas the labor illusion concerns how the visibility of service work affects customers' perceptions of a service. Neither perspective implies that making effort or service work visible necessarily improves customer evaluations in every context. Accordingly, the present review uses the effort heuristic and the labor illusion as potential explanatory lenses for understanding customer evaluations, rather than as established mechanisms underlying responses to customer-facing AI applications. This distinction is particularly important when studies do not directly examine perceived effort, the visibility of service work, or the inferential processes through which customers form evaluations.

1.2. Human and AI Involvement in Customer-facing AI Applications

Although customer evaluations constitute the primary focus of the review, understanding how the retrieved literature characterizes these evaluations also requires attention to how human and AI involvement is interpreted and represented. Accordingly, the review uses human and AI involvement as an organizing concept for understanding the roles, characteristics, and contributions of human and AI actors in customer-facing service contexts. This analytical focus is structured around two related but conceptually distinct dimensions as follows.

1.2.1. Human-likeness and Anthropomorphism

Human-likeness and anthropomorphism provide complementary perspectives for understanding how customers interpret the characteristics and capabilities of AI applications relative to human actors in customer-facing service contexts. Human-likeness refers to the extent to which an AI application is perceived to exhibit characteristics commonly associated with humans, such as warmth, competence, empathy, conversational naturalness, and social presence. Ischen et al. [13] conceptualize perceived human-likeness in text-based conversational agents along appearance-, capacity-, and personality-based dimensions, illustrating that human-likeness is multidimensional rather than a single attribute. These dimensions represent distinct aspects of perception and should not be treated as interchangeable indicators. Moreover, perceiving human-associated characteristics in an AI application does not necessarily mean that customers regard the technology as human. Research on computers as social actors demonstrates that individuals can apply social rules and expectations to technological systems, without such responses necessarily implying that they perceive computers as human [14].

Anthropomorphism refers more specifically to the attribution of human-like characteristics, motivations, intentions, or emotions to nonhuman entities. Epley, Waytz, and Cacioppo [15] explain that anthropomorphism depends not only on the characteristics of the entity but also on the perceiver's knowledge and motivations. Research on service technologies likewise indicates that anthropomorphism does not have uniformly positive or negative implications. Blut et al. [16], in a meta-analysis of service robots, chatbots, and other AI applications, show that the relationships between these applications and customer responses vary according to technological and service-context characteristics. Anthropomorphism should therefore be distinguished both from the design features of an AI application and from the customer evaluations that may follow.

1.2.2. Authenticity and Intentionality

Authenticity and intentionality provide a conceptual basis for understanding how customers interpret human and AI contributions to service outcomes. Runco [4] argues that authenticity and intentionality help distinguish human creativity from artificial creativity. Although this distinction originates in debates about creativity, it offers a useful lens for understanding how customers evaluate service outcomes that involve AI contributions.

Authenticity reflects how customers evaluate the genuineness, credibility, or legitimacy of an outcome rather than being determined solely by whether human or AI contributions were involved in its production. Grayson and Martinec [17] distinguish between indexical authenticity, which stems from a direct connection to an object's origin, and iconic authenticity, which derives from its resemblance to what is perceived as authentic. Napoli et al. [18] likewise conceptualize authenticity as a consumer-based judgment rather than an inherent property of an offering. Research on marketing communications attributed to AI further demonstrates that perceived AI authorship can influence authenticity judgments and subsequent consumer responses [19]. However, such effects are contingent on the communication context and how AI involvement is understood.

Intentionality concerns the purposes or intentions attributed to the production of an outcome. It draws attention to the distinction between an output that exhibits characteristics associated with creativity and one that is understood as reflecting a particular purpose or creative intent. Research suggests that people may attribute intentionality to AI systems, and that such attributions can influence how they judge the system's actions and responsibility [20, 21]. In customer-facing AI contexts, such perceptions may influence how customers interpret the respective contributions of human and AI actors. However, intentions attributed to a creator or production process should not be conflated with those attributed to an AI system, nor should an apparently purposeful output be taken as evidence that the system itself possesses intentions.

1.3. Conceptual Synthesis and Analytical Framing

The literature reviewed above provides the conceptual foundation for investigating what customer evaluations of customer-facing AI applications are reported in the retrieved literature and for assessing how that literature conceptualizes human and AI involvement in ways that inform the understanding of these evaluations. The concepts operate at different analytical levels, as illustrated in **Figure 1**. Artificial creativity serves as the conceptual point of departure, while generative AI, large language models, ChatGPT, and chatbots delineate the technological scope of the retrieval strategy. Within this scope, customer evaluations constitute the primary analytical focus, with trust and perceived value treated as the focal evaluative dimensions. The effort heuristic and the labor illusion serve as potential explanatory lenses for examining whether and how perceptions of production relate to customer evaluations. Human and AI involvement provides a secondary analytical perspective and serves as the organizing concept for synthesizing the retrieved literature through two analytical dimensions, one concerned with perceived human-associated characteristics, including human-likeness and anthropomorphism, and the other with production and outcome interpretation, including authenticity and intentionality. Overall, the review does not assume that these concepts were explicitly examined in every included study. The synthesis distinguishes concepts explicitly investigated by individual studies from those addressed indirectly through related findings or theoretical interpretations, without assuming that the relationships among them operate uniformly across technologies and service contexts.

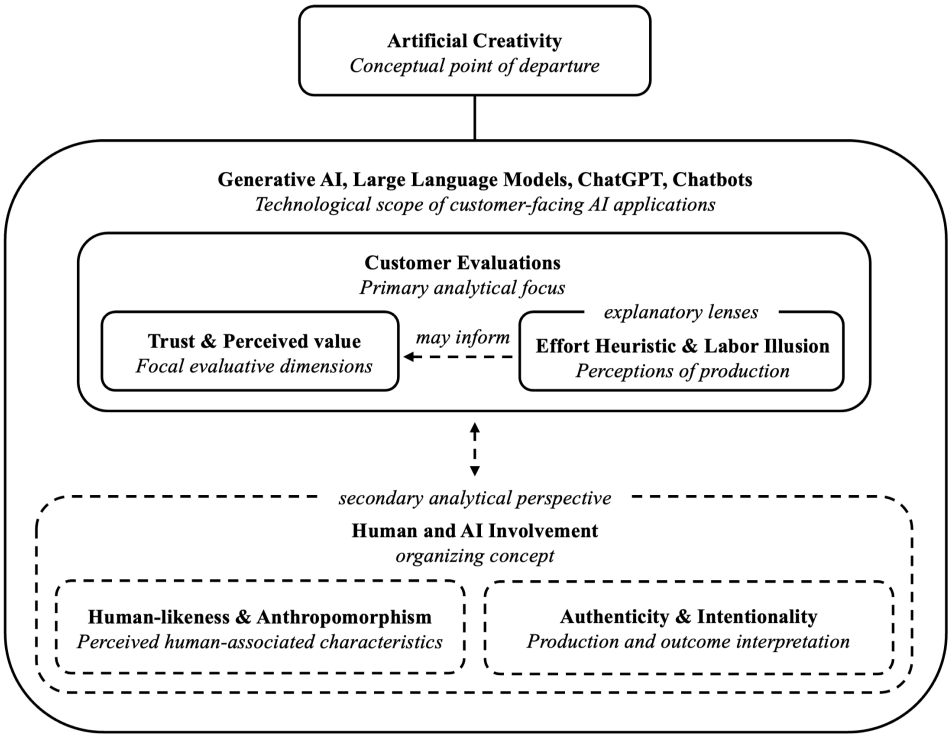


Figure 1. Analytical framing of the exploratory scoping review. Solid outlines indicate terms explicitly included in the search string, whereas dashed outlines indicate analytical concepts not explicitly included in that string. Connections show the structure of the synthesis and potential relationships discussed in the retrieved literature, rather than established causal relationships. Figure drawn by the first author.

2. Method

2.1. Scoping Review Design and Search Strategy

A scoping review approach was selected because the relevant literature spans diverse service contexts, research designs, terminology, and levels of construct operationalization. The review aimed to map the conceptual and empirical landscape of the retrieved literature, identify how relevant constructs were operationalized, and examine recurring patterns across contexts, rather than estimate a common effect or establish causal relationships. The search strategy, the database, the eligibility criteria, and the resulting corpus bounded its scope. Thus, it does not claim comprehensive coverage of research on customer-facing AI applications. The review was reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR) [22], consistent with the exploratory mapping objective [23].

A literature search was conducted in the Scopus database on 3 February 2026, restricted to publications from 2023 onward, using the following search string: ("Artificial Creativity" OR "Generative AI" OR "Large Language Models" OR "ChatGPT" OR "Chatbots") AND ("Customer Trust" OR "Perceived Value" OR "Effort Heuristic" OR "Labor

illusion")*.* The search string was structured around two categories of terms. The first category captured AI-related technologies and concepts, including *Artificial Creativity*, *Generative AI*, *Large Language Models*, *ChatGPT*, and *Chatbots*. The second category focused on customer evaluations and theoretical perspectives on how individuals assess service outcomes, including *Customer Trust*, *Perceived Value*, *Effort Heuristic*, and *Labor Illusion*. Within each category, the Boolean operator *OR* allowed for variation in terminology, while *AND* ensured that retrieved records contained at least one term from each category. The logic behind the search strategy is that artificial creativity and the associated AI technologies identify the focal phenomenon and technological mechanisms through which AI contributes to service creation and delivery. Customer trust and perceived value identify key dimensions of customer evaluation, while the effort heuristic and the labor illusion provide theoretical lenses for understanding how customers evaluate service outcomes and interpret the visibility of human effort. The search terms served as retrieval criteria and were not intended to represent an exhaustive or mandatory set of constructs examined in the retrieved literature.

2.2. Study Selection and Eligibility Criteria

The initial search yielded 154 records. After restricting the results to English-language journal articles classified as the document type Article, 100 records remained. Title screening excluded 56 records, leaving 44 for abstract screening. A further 26 records were excluded at the abstract stage, after which 18 records proceeded to full-text assessment. Of these, seven studies were excluded following full-text screening, including two that fell outside the scope of customer-facing AI service contexts, and five that did not provide sufficient substantive evidence relevant to the research questions. As a result, the final corpus comprised 11 peer-reviewed journal articles. The study selection process is illustrated in **Figure 2**.

As shown in **Table 1**, eligibility decisions considered the publication restrictions, the relevance of the AI application to a customer-facing service context, the presence of a customer, user, or service-recipient perspective, and the substantive relevance of the study to the review's interest in the interpretation of human and AI involvement. Accordingly, the final corpus should be understood as the result of both the specified database search and the subsequent relevance assessment, rather than as a sample selected exclusively based on the search string's explicit terms for customer evaluation.

2.3. Data Charting and Analytical Approach

The 11 included studies were charted to capture their service contexts, AI applications, research designs, samples, focal constructs, and reported customer outcomes. The analysis centered on the two research questions, examining how the studies reported customer evaluations and how they addressed human and AI involvement in ways relevant to those evaluations. In addition, design recommendations and governance-related considerations were charted and synthesized as supplementary findings, providing context for the subsequent discussion of practical implications.

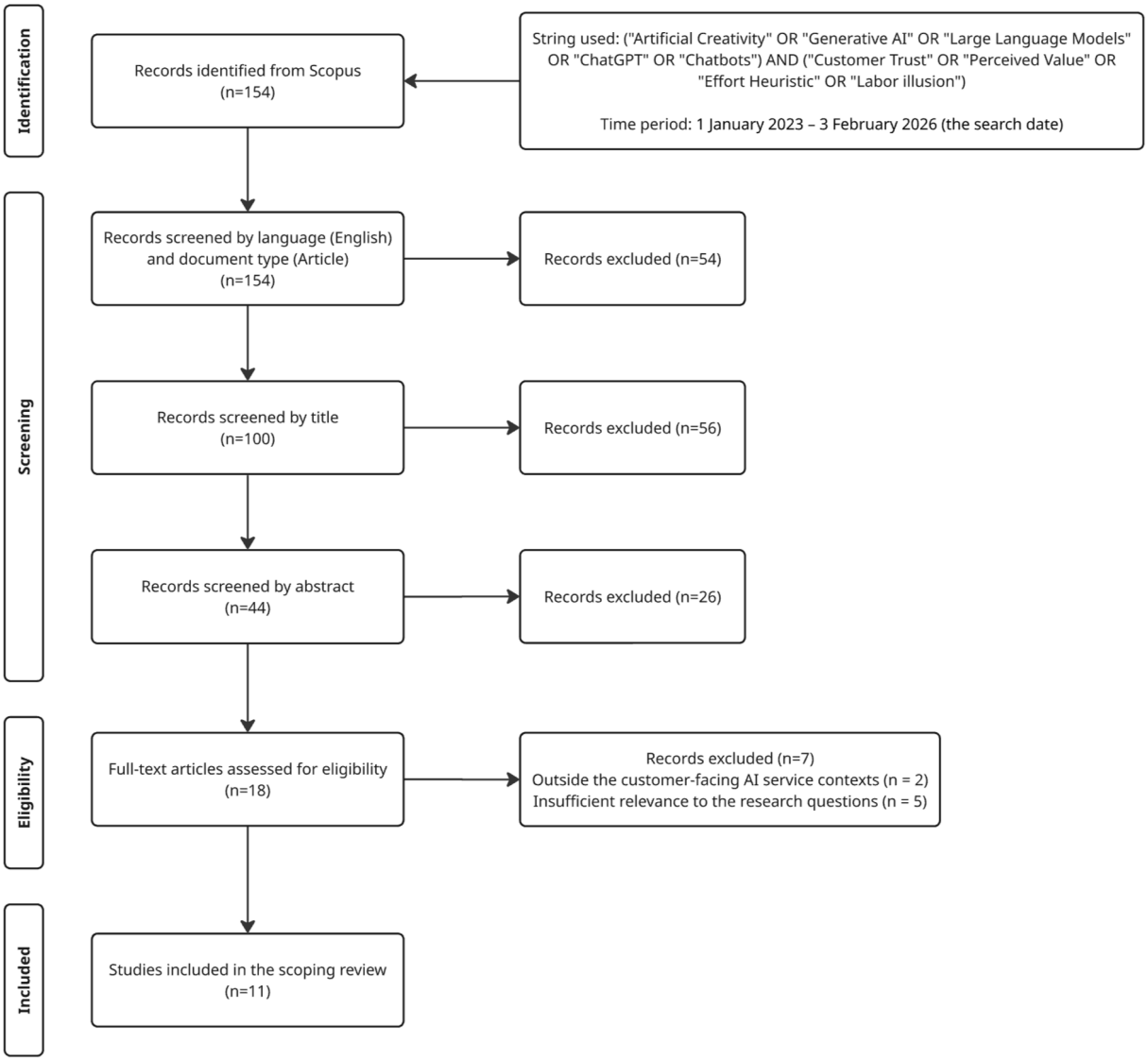


Figure 2. PRISMA-ScR flow diagram of the study selection process. Figure drawn by the first author.

Table 1. Eligibility criteria of the scoping review. Table drawn by the authors.

Criterion	Inclusion	Exclusion
Publication characteristics	English-language journal article published from 2023 onward and classified as an article in Scopus	Publication outside the selected date, language, or document-type restrictions
Application context	AI application situated in a customer-facing, service, or consumer context	Application or sector outside the scope of the review
Perspective	Focus on customers, users, or service recipients and their responses to the application	No relevant customer, user, or service-recipient perspective
Substantive relevance	Findings or conceptual discussion relevant to customer evaluations, human and AI involvement, or the review's related evaluative concerns	Insufficient substantive evidence to address the review questions

A distinction was maintained between the search terms specified in the original search string and the analytical concepts used to examine the included studies. The search terms defined the initial pool of records, whereas the analytical concepts guided the subsequent characterization and synthesis of the literature. Human-likeness, anthropomorphism, authenticity, and intentionality were considered in the analysis, although they were not predefined as search anchors in the original search string. These concepts therefore served as analytical categories for examining the corpus rather than as additional search criteria.

Given the heterogeneity of the corpus, the analysis distinguished between two dimensions of conceptual coding. The first captured how a concept was addressed: explicit terminology was recorded when a study used the focal term or a clear linguistic equivalent, whereas implicit conceptual attention was recorded when a study substantively examined a conceptually relevant issue without necessarily using that terminology. The second captured the status of the supporting evidence: whether the concept was empirically measured, discussed substantively in the study's theoretical interpretation, or merely mentioned without corresponding substantive analysis. These dimensions were coded separately because explicit use of a term did not necessarily indicate that the concept was empirically examined, while conceptually relevant evidence could occur without explicit terminology.

Table 2. Characteristics of the 11 studies included in the review. Table drawn by the first author.

Study	Service context / AI application	Research Design	Sample	Main analytical focus
Casaló et al. [24]	Generative-AI chatbot	Experiment	n = 308	Humanness, intelligence, service value
Chong et al. [25]	Embodied financial advisor	Mixed methods	n = 596	Agent role, appearance, engagement
Fu et al. [26]	Online shopping chatbot	Survey	n = 395	Human-like characteristics, trust
Guinalíu et al. [27]	Food-service imagery	Survey	n = 241	AI-generated images, value, risk
Hamedani et al. [28]	E-commerce chatbot	Survey	n = 432 (402 valid)	Anthropomorphic cues, adoption
Hao et al. [29]	Digital museum experience	Survey	n = 726	AI value, novelty, adoption
Hennighausen et al. [30]	E-commerce customer service	Experiment	n = 294	AI disclosure, trust
Jia et al. [31]	Hotel booking	Experiment	n = 764	AI- versus human-generated review summaries
Li et al. [32]	Online shopping chatbot	Experiment	n = 568	Anthropomorphism, trust, purchasing
Wang et al. [33]	Online shopping chatbot	Survey	n = 363	Anthropomorphism, trust, adoption
Yamagishi et al. [34]	Tourism information services	Survey	n = 697	AI-generated information, trust, purchasing

3. Findings

3.1. Characteristics and Conceptual Coverage of the Corpus

Table 2 summarizes the characteristics of the 11 included studies, detailing their customer-facing AI contexts, research designs, sample sizes, and principal areas of investigation. Overall, the review corpus addressed customer-facing AI through heterogeneous constructs, with human-associated characteristics among the most frequently represented. These differences in terminology and operationalization are important when interpreting findings from customer evaluations.

Specifically, the review corpus covered a range of customer-facing AI applications, including generative AI chatbots, embodied conversational agents for financial advice, online shopping chatbots, AI-generated food imagery, e-commerce customer service, digital museum experiences, AI-generated hotel review summaries, and tourism information services.

In terms of research design, the corpus included six survey studies, four experiments, and one mixed-methods study. Reported sample sizes ranged from 241 to 764 participants, with Hamedani et al. [28] reporting 432 respondents recruited and 402 valid responses. The included studies therefore varied in their applications, participants, designs, and outcomes, limiting direct comparisons across the full corpus.

Notably, the conceptual coverage of the corpus was uneven. Creativity was rarely addressed explicitly, while authenticity-related concerns and human-associated characteristics were more commonly represented. Intentionality was addressed less consistently and often indirectly, through anthropomorphic cues or distinctions between human and AI production. The concepts and terminology varied across studies and were not interchangeable. For example, Fu et al. [26] distinguished empathy and social presence from anthropomorphism and reported different relationships with trust. The following sections examine these conceptual distinctions in greater detail.

3.2. Customer Evaluations of Customer-facing AI Applications

While this review focuses on trust and perceived value, the included studies examined diverse forms of customer evaluation, including perceived risk, engagement, adoption, and purchasing-related outcomes. These dimensions are related but not interchangeable, and not all studies directly examine both trust and perceived value. Furthermore, although the effort heuristic and the labor illusion informed the search strategy and theoretical framing, neither was directly examined or tested as an explanatory mechanism for customer evaluations in the included studies. Across the review corpus, customer evaluations varied with the characteristics of the AI application and the outcomes examined. **Table 3** summarizes three broad patterns, providing an overview rather than a direct comparison of equivalent measures across studies. Some studies associated human-associated characteristics or service-related cues with more favorable evaluations, whereas others reported less favorable responses under particular AI provenance or disclosure conditions. Overall, the findings indicate heterogeneous, context-dependent customer evaluations rather than a uniform response to AI.

Table 3. Cross-study patterns in customer evaluations of customer-facing AI applications. Table drawn by the first author.

Pattern	Corpus signal	Main finding
human-associated and service-related cues	[24], [25], [28], [29], [32], [33], [34]	Human-associated characteristics and service cues were linked to trust-related, value-related, adoption, or purchase outcomes. The constructs and reported relationships varied across studies.
Different effects of human-associated cues	[26]	Empathy and social presence were associated with greater trust, while anthropomorphism itself showed a negative association under the studied conditions.
AI provenance and disclosure cues	[27], [30], [31]	AI-generated content or AI-related disclosure was associated with less favorable evaluations in some comparisons, including lower trust, lower perceived value, or higher perceived risk. Findings varied by context and outcome.

3.2.1. Trust-related Evaluations

Trust-related findings varied across AI applications, human-associated characteristics, and production conditions. Several studies reported more favorable trust responses when AI applications were perceived as useful, accurate, competent, or capable of providing relevant information. Other studies found that human-associated cues were relevant to trust, although the effects varied depending on the specific characteristic examined. For instance, Wang et al. [33] associated trust with anthropomorphism, perceived usefulness, and ease of use. Li et al. [32] reported more favorable trust responses when chatbot characteristics conveyed warmth and competence. Hamedani et al. [28] likewise linked anthropomorphic chatbot features to trust. Hao et al. [29] and Yamagishi et al. [34] emphasized information-related characteristics, including accuracy, authenticity or trustworthiness, usefulness, and relevance to customer queries. However, the results were not uniformly positive for human-associated characteristics. Fu et al. [26] found that empathy and social presence could increase trust, while anthropomorphism itself had a negative effect. This finding indicates that different characteristics associated with humanness can have different relationships with customer responses. It also cautions against treating anthropomorphism as a general predictor of trust.

Some studies compared responses across human-only, hybrid, or AI-generated communication and content. Hennighausen et al. [30] compared e-commerce service emails attributed to a human employee, a human–AI combination, or AI alone. The findings indicate the highest trust in employee-only communication and the lowest trust in the hybrid condition. Jia et al. [31] reported lower trust in AI-generated hotel review summaries than in human-generated summaries. In both cases, the source or production condition was relevant to the reported differences in trust, although the tasks and service contexts differed. However, these findings should not be interpreted as establishing a universal negative effect of AI involvement or disclosure. The studies varied in their applications, the roles assigned to AI, the content evaluated, and the conditions under which customers encountered the information.

The trust-related findings therefore reveal a context-dependent pattern: useful, accurate, relevant, or appropriately human-associated characteristics can accompany more favorable trust responses, while AI involvement in some communication or content-production conditions can accompany less favorable responses. However, the review corpus does not establish a single relationship that applies consistently across customer-facing AI applications.

3.2.2. Perceived Value and Related Evaluations

The evidence concerning perceived value was more varied in both operationalization and directness. Some studies explicitly examined service value or related value judgments, whereas others investigated usefulness, adoption, novelty, or service evaluations. These outcomes should not automatically be treated as equivalent to perceived value. For instance, Casaló et al. [24] compared chatbot configurations and found that the emotional AI configuration received higher value evaluations than the mechanical configuration, which received the lowest. The study associated this difference with humanness-related variables. Furthermore, Hao et al. [29] examined digital museum experiences featuring AI-generated content. The findings associate more favorable value evaluations with information that was useful, novel, enjoyable, semantically relevant, and adaptable to the user's context. Chong et al. [25] similarly highlighted the importance of aligning the AI agent's role and communication style with customer needs. These findings suggest that evaluations depended not only on AI's presence, but also on the qualities of the service experience and the extent to which the application met customer needs.

Interestingly, a contrasting result was reported by Guinalíu et al. [27], which examined AI-generated food imagery. The findings report lower perceived value and greater perceived risk when imagery was identified as AI-generated. The study also reported that customer experience with AI moderated some adverse responses, with more experienced users being less susceptible to certain negative evaluations.

The evidence on perceived value does not establish that human-like design or AI-generated content consistently increases or decreases value. Instead, the reported evaluations varied by application, design characteristics, information relevance, and whether the output was identified as AI-generated.

3.3. Human and AI Involvement in Relation to Customer Evaluations

The included studies addressed human and AI involvement through a range of concepts. However, these concepts were not equally represented or consistently operationalized. Human-associated characteristics were relatively prominent, whereas creativity was rarely examined as a distinct construct, and authenticity- and intentionality-related concerns were addressed in different ways across studies. **Table 4** summarizes the conceptual coverage of the review corpus. These categories are not mutually exclusive, and the presence of a concept does not necessarily mean that it was directly measured or explicitly theorized in a study.

Table 4. Analytical concepts relating to human and AI involvement in the corpus. Table drawn by the first author.

Analytical focus	Corpus signal	Main finding
Creativity	[24], [27]	Creativity was mentioned in relation to AI capabilities or imagery, but was not examined as a distinct empirical construct. The studies instead focused on human-associated characteristics, service value, perceived risk, and consumer responses to AI-generated outputs.
Human-likeness and anthropomorphism	[24], [25], [26], [28], [30], [32], [33]	Human-associated characteristics were relatively prominent, including humanness, warmth, competence, empathy, social presence, and humanoid characteristics. Their relationships with customer evaluations varied across cues and contexts.
Authenticity-related concerns	[25], [27], [30], [31], [33], [34]	Concerns related to authenticity involved credibility, trustworthiness, provenance, or comparisons between human- and AI-produced content. No common operational definition was evident across the studies.
Intentionality-related concerns	[27], [32], [33]	Intentionality was primarily indirect, appearing through perceived purpose, anthropomorphism, human-like attribution, or distinctions between human and AI production rather than through direct measures of AI intentionality.

3.3.1. Human-Like Cues and Anthropomorphic Attributions

Human-likeness and anthropomorphism were among the most recurrent concepts in the review corpus, but the nature and direction of their relationships with customer evaluations varied by characteristic and context. Seven studies in the review corpus examined anthropomorphism or related human-associated characteristics. The studies considered attributes such as warmth, competence, empathy, social presence, humanoid appearance, emotional intelligence, and human-like communication. These characteristics were linked to different customer responses. Casaló et al. [24] examined humanness, warmth, competence, and automated social presence in relation to service value. Chong et al. [25] considered the role and appearance of embodied conversational agents in financial advice, including whether the agent was positioned as a partner or a servant. Li et al. [32] examined anthropomorphic chatbot cues in relation to warmth, competence, trust, and chatbot overload. Wang et al. [33] studied humanoid attributes in relation to trust and willingness to use, while Hamedani et al. [28] examined anthropomorphic cues and adoption in e-commerce chatbots.

However, the findings do not support treating human-likeness or anthropomorphism as a single feature with a uniformly favorable relationship to customer evaluations. Fu et al. [26] reported contrasting effects, showing that empathy and social presence were associated with higher trust, whereas anthropomorphism was associated with lower trust. Chong et al. [25] also highlighted the importance of the agent's service role and interaction style, suggesting that how an AI application is positioned and communicates can matter alongside its appearance.

The studies therefore address human-likeness at different levels: some examine design features, some measure customers' perceptions of human-associated attributes, and others

examine attributions of human characteristics to AI. These distinctions are important because a human-like interface does not necessarily mean that customers attribute human mental states to the application, and neither design features nor anthropomorphic attributions automatically establish a particular customer response.

3.3.2. Authenticity and Intentionality in Human–AI Involvement

The review corpus gave greater attention to authenticity-related concerns, whereas intentionality typically appeared as an inferred rather than explicitly examined concept. Six studies in the review corpus contained authenticity or closely related authenticity concerns, yet their approaches differed. Some directly addressed authenticity, while others examined source information, credibility, trustworthiness, or the perceived human origin of content. For instance, Yamagishi et al. [34] incorporated authenticity into their analysis of AI-generated tourism information and destination trust. Guinalíu et al. [27] examined reactions to AI-generated food imagery, while Jia et al. [31] compared review summaries produced by AI and humans. These studies do not share a common operational definition of authenticity. The evidence is therefore better characterized as a set of authenticity-related concerns than as a consistent body of findings on a single measured construct. Source information and authenticity may be related, but knowing that AI-generated content is not the same as measuring whether a customer considers it genuine, credible, or legitimate.

Intentionality was less prominent, with three studies in the review corpus addressing related issues. These studies approached intentionality primarily by distinguishing between human and AI production or by identifying human-associated characteristics. Guinalíu et al. [27] connected intentionality to differences between human-created and AI-produced food imagery, including the absence of human intention and lived experience. Li et al. [32] discussed intention as a human characteristic that may be attributed to nonhuman agents through anthropomorphism, but focused empirically on chatbot characteristics and customer responses. Wang et al. [33] referred to intentions in their conceptual discussion of AI but excluded intentionality from their empirical criteria. Overall, intentionality remained a peripheral concept in these studies, appearing mainly in theoretical framing rather than as a directly examined basis for customer evaluations.

3.4. Additional Design and Governance Considerations

Although design and governance were not explicit targets of the search strategy, several included studies discussed recommendations for the design, functionality, or responsible use of customer-facing AI applications. These observations were retained as contextual findings because they help situate the reported customer evaluations within the practical decisions surrounding AI services. They were not treated as a separate analytical dimension equivalent to customer evaluations or human and AI involvement. Instead, they were recorded as a supplementary mapping of concerns and recommendations discussed in the included studies, as summarized in **Table 5**. However, the categories distinguish the general focus of the discussion rather than mutually exclusive themes, and the presence of a topic does not necessarily indicate that a study empirically evaluated a specific intervention or demonstrated its effectiveness.

Table 5. Design and governance considerations discussed in the included studies. Table drawn by the first author.

Category	Corpus signal	Main finding
Design- and functionality-oriented recommendations	[24], [25], [26], [28], [29], [32], [33], [34]	Recommendations focused on design characteristics, service functionality, conversational style, agent roles, usefulness, and adaptability to customer needs or contexts.
Responsibility-related considerations	[27], [29], [30], [31], [32], [34]	Studies discussed issues such as transparency, disclosure, privacy, bias, or ethical risks, although the specific issues and their treatment varied across studies.
Concrete legal or policy frameworks	[28], [30]	Explicit references to formal regulatory or policy frameworks were less common than broader discussions of responsible AI, transparency, or ethical considerations.

The most common practical recommendations concerned the design and functionality of customer-facing AI applications. Eight studies in the review corpus discussed aspects such as anthropomorphic cues, conversational style, agent roles, usefulness, or contextual adaptability. These recommendations generally focused on how applications could be configured to better suit customer needs and service situations. However, the recommendations varied in their emphasis, and their inclusion does not establish that any particular design choice will consistently improve customer evaluations across contexts.

Responsibility-related considerations were also present, although their treatment was less uniform. Six studies in the review corpus discussed matters such as transparency, disclosure, privacy, bias, or ethical risks. These discussions indicate that the included studies went beyond immediate customer responses to consider broader concerns associated with deploying AI in service settings. Nevertheless, the studies did not consistently address a common set of design and governance considerations.

Interestingly, explicit references to policy or regulatory frameworks were less frequent, appearing in only two included studies. Hamedani et al. [28] referred to the *Malaysia Digital Economy Blueprint* and the *National Artificial Intelligence Roadmap*, situating AI within a national strategic agenda. Hennighausen et al. [30] discussed the *European Union Artificial Intelligence Act*, particularly regarding transparency in AI use. These policy references differed both in policy scope and emphasis. The former provided strategic frameworks that established national priorities and directions for AI development. Meanwhile, the latter was linked to a formal regulatory framework that set legal requirements in specified contexts. Their limited presence suggests that, within the review corpus, practical design recommendations and broader responsibility-related concerns received more attention than specific policy frameworks.

4. Discussion

4.1. Trust and Perceived Value as Context-Dependent Customer Evaluations

In response to the first research question, the review corpus suggests that customer evaluations of customer-facing AI, particularly trust and perceived value, are context-dependent rather than uniform responses to AI use. Trust in customer-facing AI is not merely a favorable response to a system but also a question of whether trust is appropriately calibrated to the system's capabilities, limitations, and role. Research on trust in automation shows that trust develops through the interaction of user characteristics, situational conditions, and experience [35]. Experimental evidence further suggests that users may over-rely on AI advice, even when it conflicts with available contextual information [36]. Human-like cues and fluent interaction may inspire confidence, but this confidence does not necessarily reflect a system's actual capabilities and limitations [11, 37]. These perspectives position calibration as a useful interpretive concept, shifting attention from whether customers trust an AI application to whether that trust is appropriately aligned with the application's capabilities, limitations, and the consequences of relying on it. However, the corpus does not consistently establish whether reported trust is appropriately calibrated, nor whether greater trust leads to better service outcomes. Trust should therefore be considered in relation to the AI application's role, the task being performed, the consequences of error, and the availability of human intervention. A system supporting low-stakes information retrieval may warrant a different degree of trust from one providing financial advice or informing consequential decisions. The objective is consequently not to maximize trust, but to support appropriate trust in the context of use.

Perceived value raises a related but distinct question of whether an AI application offers benefits that customers consider worthwhile in a particular service context. In consumer research, perceived value is commonly understood as an overall evaluation of the benefits received relative to what is given up, and may incorporate both functional and emotional dimensions [6, 38]. In technology-mediated services, usefulness can contribute to customer evaluations, but it is not equivalent to perceived value [39]. Nor should perceived value be conflated with trust, which has been conceptualized as a distinct consideration in models of online customer behavior [40]. An application may be perceived as useful while still raising doubts about its credibility. Equally, customers may regard an application as credible without finding its contribution sufficiently valuable. The two evaluations may be related, but they should not be assumed to move together uniformly across applications or service settings.

The effort heuristic and the labor illusion offer additional theoretical lenses for understanding customer evaluations. Perceived effort can function as a cue for quality [10], while making service work visible can increase perceived value under certain conditions [9]. These ideas raise an important question for customer-facing AI services: when a system produces an answer or completes a task rapidly, do customers evaluate the outcome primarily on its usefulness, or does the apparent absence of human effort also shape their judgment? However, neither the effort heuristic nor the labor illusion was directly examined

as an explanatory mechanism in the studies included in this review. Their relevance to customer-facing AI services therefore remains an open question for future research, particularly regarding whether and under what conditions customers' perceptions of effort and the visibility of human work shape perceived value.

4.2. Human and AI Involvement as a Lens for Interpreting Customer Evaluations

The second research question concerns how the literature addresses human and AI involvement in ways relevant to customer evaluations. The present review suggests that customers evaluate AI applications not only in terms of their functional performance but also through perceptions of the human-associated characteristics, intentions, and contributions associated with their outputs. Anthropomorphism research has shown that people may apply human social expectations to nonhuman entities, including computers, without necessarily believing they are human [14, 15]. In customer-facing AI, human-like appearance, conversational style, warmth, and empathy may therefore shape expectations about how an application will behave and interact with customers. However, these characteristics should not be treated as interchangeable, nor should their presence be assumed to produce consistently favorable evaluations. The included studies report varied relationships between human-associated characteristics and customer responses, suggesting that the significance of these relationships depends on the characteristics presented and the service context in which they are encountered.

Human-like interfaces and relational cues may make interactions feel more accessible and socially engaging, while also shaping expectations about a system's understanding, competence, and agency [37]. For example, a conversational agent may appear empathetic or competent, yet the appearance of these qualities does not demonstrate the same capacities in the system as in a human service provider. Human-likeness and anthropomorphism are consequently useful but distinct concepts: the former concerns perceived resemblance to human characteristics, whereas the latter concerns the attribution of human characteristics or mental states to a nonhuman entity. The findings indicate that both have been addressed in the corpus, but not through a consistent explanatory model. Human-like cues may support favorable evaluations in some settings, whereas anthropomorphic characteristics may have different, or even unfavorable, relationships with trust in other contexts. The present review therefore highlights the importance of considering which human-associated qualities are represented, what expectations they may invite, and whether the application's actual role and performance support those expectations.

Authenticity and intentionality extend this issue beyond the representation of human characteristics to how customers may interpret human and AI contributions. Although related, authenticity and intentionality are not equivalent: an output may be regarded as credible even if customers do not attribute human intentions to the AI system, and attributing an intention to a system does not necessarily determine whether its output is considered authentic. Runco [4] discusses the relationship between authenticity and intentionality in the context of human and artificial creativity, offering a theoretical perspective for considering how these ideas may intersect. However, the corpus did not

consistently define or operationalize either concept. In particular, intentionality was rarely examined as a distinct empirical construct. These concepts should therefore be understood as complementary interpretive lenses rather than as established mechanisms that explain customer evaluations.

Prior research on AI disclosure illustrates how customer evaluations of AI involvement depend on context. ChatGPT-generated online reviews were perceived as less authentic, trustworthy, and useful than human-generated reviews [41]. By contrast, disclosing AI authorship in brand-voice communications did not necessarily reduce perceived authenticity or brand attitude [42]. These findings are not inherently contradictory, as they concern different types of content and evaluative settings. Disclosure may prompt customers to reconsider an output's credibility, the extent of human contribution, or the intentions attributed to those responsible for its production. Nevertheless, it does not determine how these aspects will be judged. Knowing that AI contributed to an outcome is not the same as judging the outcome to be inauthentic or attributing intentions to the AI system itself rather than to the people or organization behind it. Although human-like representations may invite such attributions, the present review does not establish how consistently customers make them or whether they explain differences in trust, perceived value, authenticity, or intentionality. These questions may be especially relevant when human contributions remain part of a service process but are not apparent to customers, as the extent to which such contributions shape customer evaluations remains an open question for future research.

4.3. Theoretical Contribution to Service Research

The present review's principal theoretical contribution is to connect customer evaluations to how human and AI involvement is represented and interpreted in customer-facing services. It offers a conceptual map of a bounded and heterogeneous evidence base, identifying distinctions that are important for interpreting the reported findings.

First, the review separates customer evaluation outcomes from the cues and characteristics through which AI is presented. Trust, perceived value, perceived risk, adoption, and purchase-related responses are not interchangeable measures of overall acceptance. Similarly, warmth, competence, empathy, social presence, and anthropomorphism do not represent a single, uniform design characteristic. Preserving these distinctions allows for a more precise interpretation of apparently conflicting results. It reduces the risk of treating any human-associated cue as a general solution to challenges of customer trust or perceived value.

Second, the review places human-likeness and anthropomorphism alongside authenticity and intentionality without treating them as equivalent constructs. Across the included studies, customer evaluations were more commonly examined in terms of outcomes such as trust, adoption, perceived usefulness, and value. In contrast, creativity, authenticity, and intentionality were less consistently treated as distinct empirical constructs. This uneven conceptual coverage limits what can be inferred from the reported evaluations, because favorable or unfavorable responses do not, on their own, reveal how customers interpreted the human and AI contributions behind an outcome. The review therefore identifies not

merely an inconsistency in terminology, but a limitation in the explanatory detail available across the corpus. This points to an opportunity for service research to examine not only whether a particular design cue is associated with customer evaluations, but also how customers interpret what that cue indicates about the system's contribution, capabilities, and role.

These contributions support a conceptual synthesis that considers customer evaluations alongside the representation and interpretation of human and AI involvement. The relationships among these elements should be viewed as questions for investigation rather than an empirically validated causal sequence. Given the heterogeneity of the included studies and the uneven operationalization of relevant constructs, the review's contribution is to clarify distinctions and identify questions that warrant more direct empirical investigation, rather than to establish a common explanatory model across customer-facing AI applications.

4.4. Practical Implications and Conditional Design Propositions

The findings on additional design and governance considerations provide a practical extension to the present review's central argument, which holds that customer evaluations cannot be understood solely by the characteristics of an AI interface or the quality of its outputs. They also depend on how the service represents AI involvement, communicates the roles of human and technical actors, and makes responsibility understandable. The included studies frequently offered recommendations concerning system design and functionality, while responsibility-related concerns varied across studies, and explicit references to legal or policy frameworks received comparatively limited attention. Notably, this pattern should be interpreted with caution, as it does not necessarily imply a general blind spot in the wider field. Rather, it suggests that, within the review corpus, practical questions of AI service configuration received more consistent attention than questions concerning how contribution and accountability should be made legible to customers. This sheds light on broader design and governance implications, highlighting the need to move beyond interface-level features to consider how human and AI contributions are organized, reviewed, and made understandable within the overall service configuration.

First, one implication is to align human-like representation with the system's capabilities. Conversational style, empathy, and relational cues may improve accessibility, yet they may also create expectations about competence or judgment that the system cannot fulfill. Design choices should therefore be evaluated in relation to the task, user needs, and consequences of misplaced expectations, rather than assuming that greater human-likeness is inherently beneficial.

Second, AI disclosure should be considered in context rather than treated as uniformly beneficial or detrimental. Communicating that AI contributed to an outcome does not necessarily clarify whether a human reviewed it or who is responsible for errors and follow-up. Where these distinctions matter, service interfaces and operational workflows can make

human review, escalation, and accountability more explicit. This requires attention to the wider service configuration, not only to interface-level disclosure.

Third, the tension between seamlessness and legibility suggests a role for productive discomfort as a design lens. Drawing on critical and speculative design [43, 44], designers might introduce proportionate interruptions or reflective cues when customers need to recognize AI involvement, uncertainty, or human oversight. Such interventions should not create friction for its own sake. Their potential value depends on whether the risk of misunderstanding justifies the additional interactional cost.

Table 6 presents the conditional design propositions that translate these implications into context-sensitive considerations and questions for future empirical investigation. Of particular note, these propositions are derived from the review's synthesis and broader theoretical framing rather than representing empirically validated design rules.

Table 6. Conditional design propositions for customer-facing AI applications. Table drawn by the first author.

Conditional proposition	Possible user attribution	Trust calibration	Relevant boundary conditions
Calibrate human-like cues	The system may be perceived as more capable, understanding, or socially present	Could support engagement, but might also invite expectations that exceed actual capability	Task, actual system capability, user goals, and consequences of error
Make AI involvement legible	The customer may infer more about the source of the output, but not necessarily about who reviewed it	Disclosure may alter trust or value evaluations; the appropriate response may depend on the task and what the user already expects	Output type, service context, customer expectations, and task significance
Clarify human contribution and responsibility	The customer can better understand where human judgment and follow-up enter the service	May support appropriate reliance and make responsibility more legible	Degree of human–AI integration, error consequences, and whether review is substantive
Use productive discomfort selectively	The customer is prompted to reconsider the output or the roles involved	May help correct consequential misunderstandings, but may also add unnecessary friction	Likelihood and consequences of misunderstanding versus interaction cost

4.5. Limitations and Directions for Future Research

The present review has several limitations that should be considered when interpreting its findings. First, the synthesis is based on a bounded set of 11 studies retrieved from a single database. Given the limited scope of the retrieval strategy and the considerable variation in research designs, conceptual definitions, and measurement approaches across the included studies, the synthesis should be understood as an exploratory mapping rather than a comprehensive account of customer responses to customer-facing AI applications. Its contribution lies in clarifying how customer evaluations and human-AI involvement are

addressed within the corpus, identifying conceptual distinctions, and highlighting areas for further empirical investigation.

Second, the included studies differ in their service contexts, AI applications, research designs, sample sizes, and outcome measures. This heterogeneity limited direct comparison and precluded conclusions about pooled effects. Although the present review distinguishes direct empirical findings from indirect connections and review-level interpretations, these differences limit the extent to which findings can be compared or combined. In particular, the reported associations do not establish a general effect of AI involvement, human-like characteristics, or disclosure on customer evaluations. The proposed relationships should consequently be treated as context-dependent interpretations and questions for further investigation, rather than as established causal explanations.

Third, the conceptual coverage of the corpus constrains the conclusions that can be drawn about how customers interpret human and AI involvement. Human-associated characteristics received comparatively more attention than creativity, authenticity, and intentionality as distinct empirical constructs. The limited and inconsistent treatment of these concepts does not necessarily indicate that they are unimportant to customers. Rather, it means that the available studies provide an uneven basis for explaining how customers attribute capability, agency, or intention to AI systems and how these attributions relate to their evaluations. The review's conceptual synthesis therefore identifies relationships that warrant investigation, but cannot determine the psychological mechanisms underlying them. In addition, the search strategy and screening process shaped the scope of the corpus, including its expanded attention to human-likeness, anthropomorphism, authenticity, and intentionality. The resulting emphasis should not be interpreted as an unbiased representation of every dimension relevant to customer evaluation.

Future research should test the proposed relationships across service contexts that differ in task purpose, perceived risk, customer expectations, and the consequences of error. It should distinguish human-like cues from information about AI involvement and from the visibility of human review, escalation, and accountability mechanisms. Trust and perceived value should also be measured directly, alongside relevant attributions, rather than inferred from adjacent outcomes or used as proxies for one another. Moreover, research-through-design offers a suitable approach for examining the conditional design propositions through the development and evaluation of design interventions in context [45]. Comparative service prototypes could vary how AI involvement, human review, and escalation are communicated, enabling researchers to examine how these configurations shape customers' interpretations of agency and responsibility. Evaluation should extend beyond preference or acceptance to assess whether designs support appropriately calibrated expectations of AI capabilities, clarify human contributions, and make accountability understandable. Such work would help identify not only whether particular design approaches affect customer evaluations, but also the conditions under which they are most appropriate.

Conclusion

Across the 11 included studies, this exploratory scoping review provides a bounded synthesis of a heterogeneous evidence base. It identifies conceptual distinctions that help clarify the relationships among customer evaluations, AI application characteristics, and perceptions of human and AI involvement. The findings indicate that customer evaluations are context-dependent rather than uniform responses to AI applications. Trust and perceived value capture distinct dimensions of customer evaluation and should not be treated as interchangeable indicators of favorable responses. Human-associated characteristics were associated with more favorable responses in some studies, but they were not consistently beneficial. Likewise, AI-generated content and disclosure of AI involvement were associated with less favorable responses in some contexts. Nonetheless, the evidence does not support a universal negative effect of AI involvement. The relevance of these characteristics depends on the application, the service context, the outcome being evaluated, and the expectations customers bring to the interaction.

The review also highlights the importance of distinguishing customer evaluations from the ways human and AI involvement is represented and interpreted. Human-likeness and anthropomorphism received comparatively more attention than creativity, authenticity, and intentionality as distinct empirical constructs. Although these concepts offer related perspectives on how customers understand AI applications and their outputs, the included studies do not establish them as a unified explanatory mechanism. In particular, favorable or unfavorable evaluations cannot, on their own, reveal whether customers attribute human-like intentions to an AI application or how they interpret the respective contributions of AI systems, service providers, and human participants in the service process. The review's conceptual contribution lies in bringing these perspectives into a common analytical view while preserving their distinctions. It therefore provides a conceptual map of a bounded and heterogeneous evidence base, rather than establishing a causal model of customer evaluations.

Building on the synthesis, the review develops four conditional design propositions: calibrating human-like cues to the system's actual capabilities, making AI involvement legible, clarifying human contributions and responsibility, and selectively using productive discomfort. These propositions are intended as starting points for investigating how these choices may shape customer evaluations under different conditions. Their value lies in making these relationships more explicit as questions for empirical investigation, rather than prescribing universally effective design rules. Future research could test the propositions across different service contexts, distinguish direct measures of trust and perceived value from related outcomes, and examine more explicitly how customers understand human and AI contributions to service outputs.

Author Contributions:

Hong Li contributed to the conceptualization of the study, theoretical framing, development of research arguments, interpretation and synthesis of findings, and manuscript writing, restructuring, and revision. Bianka Frøyland Brueland contributed to data collection, preliminary analysis, study development, preparation of the initial draft, and manuscript review. All authors reviewed and approved the final manuscript.

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Cultural and Bias-free Statement:

The authors declare that this manuscript has been prepared with attention to inclusivity and has made every effort to use respectful, non-discriminatory, and inclusive language.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process:

During the preparation of this work, the authors used Grammarly and Microsoft Copilot in order to assist with language editing, proofreading, and improving the text's clarity and readability. All intellectual content, interpretations, and scholarly judgments were developed by the authors. The authors critically reviewed and edited all AI-assisted suggestions as needed and take full responsibility for the content of the publication.

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References

1 Huang, Ming-Hui, and Roland T. Rust. "A Strategic Framework for Artificial Intelligence in Marketing." *Journal of the Academy of Marketing Science* 49, no. 1 (2021): 30–50. [CrossRef]

2 Dwivedi, Yogesh K., Nir Kshetri, Laurie Hughes, et al. "So What if ChatGPT Wrote It? Multidisciplinary Perspectives on Opportunities, Challenges and Implications of Generative Conversational AI for Research, Practice and Policy." *International journal of information management* 71 (2023): 102642. [CrossRef]

3 Runco, Mark A., and Garrett J. Jaeger. "The Standard Definition of Creativity." *Creativity Research Journal* 24, no. 1 (2012): 92–96. [CrossRef]

4 Runco, Mark A. "Updating the Standard Definition of Creativity to Account for the Artificial Creativity of AI." *Creativity Research Journal* 37, no. 1 (2025): 1–5. [CrossRef]

5 McKnight, D. Harrison, Vivek Choudhury, and Charles Kacmar. "Developing and Validating Trust Measures for e-Commerce: An Integrative Typology." *Information Systems Research* 13, no. 3 (2002): 334–359. [CrossRef]

6 Zeithaml, Valarie A. "Consumer Perceptions of Price, Quality, and Value: A Means-End Model and Synthesis of Evidence." *Journal of Marketing* 52, no. 3 (1988): 2–22. [CrossRef]

7 Dellermann, Dominik, Philipp Ebel, Matthias Söllner, and Jan Marco Leimeister. "Hybrid Intelligence." *Business & Information Systems Engineering* 61, no. 5 (2019): 637-643. [CrossRef]

8 Jarrahi, Mohammad H. "Artificial Intelligence and the Future of Work: Human-AI Symbiosis in Organizational Decision Making." *Business Horizons* 61, no. 4 (2018): 577-586. [CrossRef]

9 Buell, Ryan W., and Michael I. Norton. "The labor illusion: How operational transparency increases perceived value." *Management Science* 57, no. 9 (2011): 1564-1579. [CrossRef]

10 Kruger, Justin, Derek Wirtz, Leaf Van Boven, and T. William. Altermatt. "The Effort Heuristic." *Journal of Experimental Social Psychology 40*, no. 1 (2004): 91–98. [CrossRef]

11 Glikson, Ella, and Anita W. Woolley. "Human Trust in Artificial Intelligence: Review of Empirical Research." *Academy of Management Annals* 14, no. 2 (2020): 627-660. [CrossRef]

12 Akdim, Khaoula, and Luis V. Casaló. "Perceived Value of AI-Based Recommendation Service: The Case of Voice Assistants." *Service Business* 17 (2023): 81–112. [CrossRef]

13 Ischen, Carolin, Elizabeth Wang, and Edith G. Smit. "The Right Kind of Human-Like: Measuring and Testing Human-Likeness Perceptions of Text-Based Conversational Agents." *Human-Machine Communication* 13 (2026): 45–73. [CrossRef]

14 Nass, Clifford, and Youngme Moon. "Machines and Mindlessness: Social Responses to Computers." *Journal of Social Issues 56*, no. 1 (2000): 81–103. [CrossRef]

15 Epley, Nicholas, Adam Waytz, and John T. Cacioppo. "On Seeing Human: A Three-Factor Theory of Anthropomorphism." *Psychological Review* 114, no. 4 (2007): 864–886. [CrossRef]

16 Blut, Markus, Cheng Wang, Nancy V. Wünderlich, and Christian Brock. "Understanding Anthropomorphism in Service Provision: A Meta-Analysis of Physical Robots, Chatbots, and Other AI." *Journal of the Academy of Marketing Science* 49, no. 4 (2021): 632–658. [CrossRef]

17 Grayson, Kent, and Radan Martinec. "Consumer Perceptions of Iconicity and Indexicality and Their Influence on Assessments of Authentic Market Offerings." *Journal of Consumer Research* 31, no. 2 (2004): 296–312. [CrossRef]

18 Napoli, Julie, Sonia J. Dickinson, Michael B. Beverland, and Francis Farrelly. "Measuring Consumer-Based Brand Authenticity." *Journal of Business Research* 67, no. 6 (2014): 1090–1098. [CrossRef]

19 Kirk, Colleen P., and Julian Givi. "The AI-Authorship Effect: Understanding Authenticity, Moral Disgust, and Consumer Responses to AI-Generated Marketing Communications." *Journal of Business Research* 186 (2025): 114984. [CrossRef]

20 Shank, Daniel B., and Alyssa DeSanti. "Attributions of Morality and Mind to Artificial Intelligence after Real-World Moral Violations." *Computers in Human Behavior 86* (2018): 401-411. [CrossRef]

21 Ayad, Reem, and Jason E. Plaks. "Attributions of Intent and Moral Responsibility to AI Agents." *Computers in Human Behavior: Artificial Humans* 3 (2025): 100107. [CrossRef]

22 Tricco, Andrea C., Erin Lillie, Wasifa Zarin, et al. "PRISMA Extension for Scoping Reviews (PRISMA-ScR): Checklist and Explanation." *Annals of Internal Medicine* 169, no. 7 (2018): 467–473. [CrossRef]

23 Munn, Zachary, Danielle Pollock, Hanan Khalil, et al. "What are scoping reviews? Providing a formal definition of scoping reviews as a type of evidence synthesis." *JBI evidence synthesis* 20, no. 4 (2022): 950-952. [CrossRef]

24 Casaló, Luis V., Paola Millastre-Valencia, David Belanche, and Carlos Flavián. "Intelligence and Humanness as Key Drivers of Service Value in Generative AI Chatbots." *International Journal of Hospitality Management* 128 (2025): 104130. [CrossRef]

25 Chong, Terrence, Ting Yu, Debbie Isobel Keeling, Ko de Ruyter, and Tim Hilken. "Stakeholder Engagement with AI Service Interactions." *Journal of Product Innovation Management* 43, no. 1 (2026): 31–56. [CrossRef]

26 Fu, Jindi, Samar Mouakket, and Yuan Sun. "The Role of Chatbots' Human-Like Characteristics in Online Shopping." *Electronic Commerce Research and Applications* 61 (2023): 101304. [CrossRef]

27 Guinalíu, Miguel, Daniel Belanche, and Luis V. Casaló. "The Influence of AI-Generated versus Real Food Images on Perceived Value and Negative WOM." *British Food Journal* (2026): 1–24. [CrossRef]

28 Hamedani, Sharareh S., Yuvanraaj Manivannan, Lam K. Yoke, Lim S. Ee, and Sarfraz Aslam. "Unveiling the Impact of AI-Chatbot Attributes and Anthropomorphic Cues in E-Commerce." *Pakistan Journal of Commerce and Social Sciences* 19, no. 3 (2025): 441–467. [CrossRef]

29 Hao, Xiaoyan, Junping Xu, and Ying Wang. "How Generative AI Shapes User Perceived Value and Adoption Intention in Digital Museum Experiences." *Heritage Science* 13, no. 1 (2025): 608. [CrossRef]

30 Hennighausen, Christine, Vanessa G. Y. Navarro-Schär, and Eric Eller. "AI-Mediated Communication in E-Commerce: Implications for Customer Trust." *International Journal of Consumer Studies* 49, no. 5 (2025): e70111. [CrossRef]

31 Jia, Shizhen, Oscar H. Chi, and Christina G. Chi. "Unpacking the Impact of AI vs. Human-Generated Review Summary on Hotel Booking Intentions." *International Journal of Hospitality Management 126* (2025): 104030. [CrossRef]

32 Li, Yang, Zhenghua Gan, and Bowen Zheng. "How Do Artificial Intelligence Chatbots Affect Customer Purchase? Uncovering the Dual Pathways of Anthropomorphism on Service Evaluation." *Information Systems Frontiers* 27, no. 1 (2025): 283–300. [CrossRef]

33 Wang, Youhua, Muhammad Mohsin, and Khalid Jamil. "Customer Trust and Willingness to Use Shopping Assistant Humanoid Chatbot." *The Service Industries Journal* 45, no. 15–16 (2025): 1348–1372. [CrossRef]

34 Yamagishi, Kafferine, Charis A. Bitong, and Maria C. Badilla. "How Does AI Chatbot-Generated Information Influence Destination Trust and Purchase Decisions in Tourism?" *Journal of Hospitality and Tourism Insights* 9, no. 5 (2026): 1926–1947. [CrossRef]

35 Hoff, Kevin A., and Masooda Bashir. "Trust in Automation: Integrating Empirical Evidence on Factors That Influence Trust." *Human Factors* 57, no. 3 (2015): 407–434. [CrossRef]

36 Klingbeil, Artur, Cassandra Grützner, and Philipp Schreck. "Trust and Reliance on AI: An Experimental Study on the Extent and Costs of Overreliance on AI." *Computers in Human Behavior* 160 (2024): 108352. [CrossRef]

37 Sundar, S. Shyam. "Rise of Machine Agency: A Framework for Studying the Psychology of Human-AI Interaction (HAII)." *Journal of Computer-Mediated Communication* 25, no. 1 (2020): 74-88. [CrossRef]

38 Sweeney, Jillian C., and Geoffrey N. Soutar. "Consumer Perceived Value: The Development of a Multiple Item Scale." *Journal of Retailing* 77, no. 2 (2001): 203–220. [CrossRef]

39 Davis, Fred D. "Perceived Usefulness, Perceived Ease of Use, and User Acceptance of Information Technology." *MIS Quarterly* 13, no. 3 (1989): 319–340. [CrossRef]

40 Gefen, David, Elena Karahanna, and Detmar W. Straub. "Trust and TAM in Online Shopping: An Integrated Model." *MIS Quarterly* 27, no. 1 (2003): 51–90. [CrossRef]

41 Amos, Clinton, and Lixuan Zhang. "Consumer Reactions to Perceived Undisclosed ChatGPT Usage in an Online Review Context." *Telematics and Informatics* 93 (2024): 102163. [CrossRef]

42 Kirkby, Alexandra, Carsten Baumgarth, and Jörg Henseler. "To Disclose or Not Disclose, Is No Longer the Question: Effect of AI-Disclosed Brand Voice on Brand Authenticity and Attitude." *Journal of Product & Brand Management* 32, no. 7 (2023): 1108–1122. [CrossRef]

43 DiSalvo, Carl. *Adversarial Design*. Cambridge, MA: MIT Press, 2012. [CrossRef]

44 Dunne, Anthony, and Fiona Raby. *Speculative Everything: Design, Fiction, and Social Dreaming*. Cambridge, MA: MIT Press, 2013.

45 Zimmerman, John, Jodi Forlizzi, and Shelley Evenson. "Research through Design as a Method for Interaction Design Research in HCI." *In Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 493–502. New York: Association for Computing Machinery, 2007. [CrossRef]